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IOT OT SECURITY

MODULE 18

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1: IOT AND OT SECURITY

1: IoT AND OT SECURITY

1.1 INTRODUCTION

The rapid growth of **Internet of Things (IoT)** and **Operational Technology (OT)** systems has transformed modern industries, homes, and critical infrastructure. While these technologies improve efficiency and automation, they also introduce **significant security risks** due to poor design, limited security controls, and large attack surfaces.

IoT and OT systems are increasingly targeted by attackers because they often:

- Operate continuously
 - Control physical processes
 - Lack strong authentication and encryption
 - Are difficult to patch
-

1.2 UNDERSTANDING INTERNET OF THINGS (IOT)

The **Internet of Things (IoT)** refers to a network of physical devices that:

- Collect data using sensors
- Communicate over the internet or local networks
- Perform automated actions

Examples of IoT Devices

- Smart home devices (cameras, lights, thermostats)
 - Wearable devices
 - Smart TVs and appliances
 - Industrial sensors
 - Healthcare monitoring devices
-

1.3 UNDERSTANDING OPERATIONAL TECHNOLOGY (OT)

Operational Technology (OT) refers to systems that:

1: IOT AND OT SECURITY

- Monitor and control physical processes
- Are used in industrial environments
- Operate critical infrastructure

Examples of OT Systems

- Industrial Control Systems (ICS)
 - SCADA systems
 - PLCs (Programmable Logic Controllers)
 - Power grids
 - Manufacturing systems
-

1.4 DIFFERENCE BETWEEN IOT AND OT

Aspect	IoT	OT
Purpose	Data collection & automation	Control of physical processes
Environment	Consumer & enterprise	Industrial & critical infrastructure
Downtime tolerance	Medium	Very low
Security focus	Often weak	Traditionally isolated

Despite differences, **both are now interconnected**, increasing risk.

1.5 WHY IOT AND OT SECURITY IS CRITICAL

Security is critical because:

- Compromise can cause **physical damage**
 - Attacks can disrupt essential services
 - Safety of people may be at risk
 - Recovery is slow and expensive
-

1: IOT AND OT SECURITY

Unlike IT systems, failures may affect **human lives and national security**.

1.6 EVOLUTION OF IOT AND OT ATTACKS

Earlier OT systems were:

- Isolated
- Proprietary
- Not internet-connected

Modern environments now use:

- IP-based communication
- Cloud integration
- Remote access

This convergence has made them attractive attack targets.

1.7 COMMON CHARACTERISTICS OF IOT AND OT SYSTEMS

- Long device lifecycles
- Limited computing resources
- Legacy protocols
- Minimal built-in security
- Hardcoded credentials
- Rare updates

These characteristics increase vulnerability.

1.8 THREAT LANDSCAPE FOR IOT AND OT

Threat actors target IoT and OT for:

- Espionage

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- Sabotage
- Financial gain
- Disruption of services

Both cybercriminals and nation-state actors are involved.

1.9 ROLE OF ETHICAL HACKING IN IOT AND OT

Ethical hackers help by:

- Identifying vulnerabilities
- Assessing risk exposure
- Improving resilience
- Supporting regulatory compliance

Testing must be **carefully controlled** due to safety risks.

1.10 LEGAL AND ETHICAL CONSIDERATIONS

Testing IoT and OT systems:

- Requires explicit authorization
- Must avoid service disruption
- Must follow safety guidelines

Unauthorized testing can cause serious consequences.

2 – IOT AND OT ARCHITECTURE AND COMPONENTS

2 – IoT AND OT ARCHITECTURE AND COMPONENTS

2.1 IOT ARCHITECTURE OVERVIEW

IoT architecture typically consists of multiple layers that enable data collection, processing, and action.

2.1.1 Perception Layer

- Sensors and actuators
- Collect physical data (temperature, motion, pressure)

2.1.2 Transport Layer

- Transfers data using protocols (Wi-Fi, Bluetooth, cellular)
- Often insecure or lightly protected

2.1.3 Processing Layer

- Cloud platforms or edge computing
- Data storage and analytics

2.1.4 Application Layer

- User-facing applications
 - Dashboards and automation logic
-

2.2 OT ARCHITECTURE OVERVIEW

OT architecture focuses on **real-time control of physical processes**.

2.2.1 Field Devices

- Sensors, actuators, PLCs
- Directly interact with physical equipment

2.2.2 Control Layer

- PLCs, RTUs, DCS
- Executes control logic

2.2.3 Supervisory Layer

- SCADA systems

2 – IOT AND OT ARCHITECTURE AND COMPONENTS

- Monitoring and control interfaces

2.2.4 Enterprise Integration Layer

- Connection to IT systems
 - Increases exposure to cyber threats
-

2.3 CONVERGENCE OF IT, IOT, AND OT

Modern environments combine:

- Traditional IT networks
- IoT smart devices
- OT industrial systems

This convergence increases:

- Attack surface
 - Complexity
 - Risk of lateral movement
-

2.4 SUMMARY OF PART 2

Part 2 explained:

- IoT layered architecture
- OT control architecture
- How convergence introduces security risks

3: IOT AND OT THREAT LANDSCAPE

3: IoT AND OT THREAT LANDSCAPE

3.1 INTRODUCTION TO THREAT LANDSCAPE

IoT and OT systems face threats that differ significantly from traditional IT systems due to their physical impact and operational constraints.

3.2 COMMON THREAT ACTORS

- Cybercriminals
 - Hacktivists
 - Nation-state actors
 - Insider threats
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3.3 COMMON IOT THREATS

- Unauthorized device access
 - Data interception
 - Device hijacking
 - Botnet recruitment
 - Privacy invasion
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3.4 COMMON OT THREATS

- Process manipulation
- System sabotage
- Production disruption
- Safety system compromise

3: IOT AND OT THREAT LANDSCAPE

3.5 MOTIVATION BEHIND ATTACKS

- Financial gain
- Espionage
- Political influence
- Infrastructure disruption

4: IOT AND OT VULNERABILITIES

4: IoT AND OT VULNERABILITIES

4.1 COMMON IOT VULNERABILITIES

- Hardcoded credentials
 - Weak authentication
 - Insecure firmware
 - Lack of encryption
 - Insecure update mechanisms
-

4.2 COMMON OT VULNERABILITIES

- Legacy protocols without security
 - Flat network architecture
 - Lack of monitoring
 - Unpatched systems
 - Physical access weaknesses
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4.3 SUPPLY CHAIN RISKS

- Compromised firmware
 - Third-party components
 - Vendor dependency
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4.4 HUMAN AND ORGANIZATIONAL FACTORS

- Lack of security awareness
- Poor asset management
- Misconfiguration

5 – IOT AND OT ATTACK SURFACES

5 – IoT AND OT ATTACK SURFACES

5.1 NETWORK-BASED ATTACK SURFACE

- Wireless communication channels
 - Remote access interfaces
 - Cloud connectivity
-

5.2 DEVICE-BASED ATTACK SURFACE

- Firmware
 - Debug interfaces
 - Physical ports
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5.3 APPLICATION AND API ATTACK SURFACE

- Web dashboards
 - Mobile applications
 - Cloud APIs
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5.4 PHYSICAL ATTACK SURFACE

- Direct device access
- Tampering
- Environmental exposure

6: IMPACT OF IOT AND OT COMPROMISE

6: IMPACT OF IoT AND OT COMPROMISE

6.1 IMPACT ON SAFETY

- Risk to human life
 - Equipment damage
 - Environmental hazards
-

6.2 IMPACT ON AVAILABILITY

- Production downtime
 - Service disruption
 - Infrastructure failure
-

6.3 IMPACT ON CONFIDENTIALITY AND INTEGRITY

- Data leakage
 - Manipulated sensor data
 - Loss of trust in systems
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6.4 BUSINESS AND NATIONAL IMPACT

- Financial loss
- Regulatory penalties
- National security threats

7: DETECTION AND MONITORING STRATEGIES

7: DETECTION AND MONITORING STRATEGIES

7.1 CHALLENGES IN IOT AND OT MONITORING

- Limited device visibility
 - Legacy protocols
 - Performance constraints
-

7.2 NETWORK MONITORING

- Traffic anomaly detection
 - Baseline behavior analysis
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7.3 DEVICE AND SYSTEM MONITORING

- Firmware integrity checks
 - Configuration monitoring
-

7.4 CENTRALIZED LOGGING AND SIEM

- Correlation of OT and IT events
- Early warning indicators

8: PREVENTION AND MITIGATION STRATEGIES

8: PREVENTION AND MITIGATION STRATEGIES

8.1 SECURE DESIGN PRINCIPLES

- Security by design
 - Defense in depth
 - Zero trust concepts
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8.2 NETWORK SEGMENTATION

- Separation of IT, IoT, and OT
 - Controlled communication paths
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8.3 SECURE CONFIGURATION AND PATCH MANAGEMENT

- Firmware updates
 - Configuration hardening
-

8.4 ACCESS CONTROL AND AUTHENTICATION

- Strong authentication
 - Role-based access control
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8.5 INCIDENT RESPONSE FOR IOT AND OT

- Safety-first response
- Coordinated IT–OT teams

9: ROLE OF ETHICAL HACKING IN IoT AND OT

9.1 IMPORTANCE OF ETHICAL HACKING

- Identifies weaknesses safely
 - Improves resilience
 - Supports compliance
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9.2 CONSTRAINTS IN TESTING

- Safety risks
 - Availability requirements
 - Regulatory limitations
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9.3 ETHICAL HACKING OUTCOMES

- Risk prioritization
- Improved architecture
- Incident preparedness

10: CONCLUSION AND REFERENCES

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10.1 CONCLUSION

IoT and OT systems have become essential to modern life and critical infrastructure. However, their unique characteristics—long lifecycles, limited security, and physical impact—make them high-value targets for attackers.

Understanding vulnerabilities, attack surfaces, and defensive strategies from a **theoretical and ethical perspective** is crucial for securing these environments.

10.2 KEY TAKEAWAYS

- IoT and OT attacks can cause physical damage
 - Convergence increases risk
 - Prevention is more effective than response
 - Security must be continuous
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10.3 REFERENCES

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3. OWASP – *IoT Top 10*
4. IEC 62443 – *Industrial Automation Security*
5. ISO/IEC 27001 – *Information Security Management*